

External inventory changes except for the hotbar are not registered in a creative mode item selection screen

MC-268431 Inventory

Status	Resolved / Fixed
Affected	1.20.4, 24w06a, 1.21, 24w33a
Fixed in	24w38a
Reported	2024-02-09

STEPS TO REPRODUCE

1. Create or enter a Creative world with default settings.
2. Fill the hotbar with items.
3. Open the Creative inventory.
4. Select a Creative inventory tab other than the Survival Inventory tab, such as the Search tab.
5. While keeping the Creative inventory open on that tab, take another item and drop it onto the ground.
6. Wait until the dropped item is picked up.
7. Observe that the picked-up item does not appear in the inventory while the Creative inventory remains open.

OBSERVED BEHAVIOUR

The item is picked up by the server while the Creative inventory remains open on a non-Survival tab, but the client does not apply the corresponding slot update for non-hotbar inventory slots. As a result, the item is not displayed in the player inventory, even though it is present in the server-side inventory. The hotbar continues to update normally. Executing `/clear @s <item>` confirms the desynchronization by reporting the item that the client failed to display.

EXPECTED BEHAVIOUR

While the Creative inventory is open on any non-Survival tab, picking up a dropped item should update the client's inventory immediately. The item should appear in an available non-hotbar inventory slot and remain visible when the inventory is viewed, with the client accurately reflecting the server's inventory contents and avoiding any desynchronization.

ENVIRONMENT

Minecraft Java Edition Versions: 1.20.4, 24w06a, 1.21, 24w33a (includes snapshots)

ORIGINAL DESCRIPTION

External inventory changes (e.g. by picking up items) are not registered on the client if you are in creative mode inventory and are not in the Survival Inventory tab, causing a desync.

Steps to reproduce:

- Fill your hotbar with items.
- Go into the creative mode inventory and go to any tab, but the inventory tab (e.g. the search tab).
- While in the tab, take any other item and throw it onto the ground.
- Wait until you pick it up and observe whether or not the item appears in the inventory.

Expected behavior:

The item is picked up and should show up in the inventory.

Actual behavior:

The item does not appear in the inventory. However, if you try to remove that particular item, e.g. using `/clear @s <item>`, the server will report that the item indeed was in the inventory.

Code analysis:

```
net.minecraft.client.multiplayer.ClientPacketListener
public void handleContainerSetSlot(ClientboundContainerSetSlotPacket packet) {
```

```
// ...
boolean conditional = false; // Line 1200
if (minecraft.screen instanceof CreativeModelInventoryScreen x) { // Line 1202
conditional = !x.isInventoryOpen(); // Line 1203
}
// 0 means player inventory
if (packet.getContainerId() == 0 && InventoryMenu.isHotbarSlot(slotId)) {
// ...
player.inventoryMenu.setItem(slotId, packet.getStateId(), item);
} else if (packet.getContainerId() == player.containerMenu.containerId && (packet.getContainerId() != 0 ||
!conditional)) {
player.containerMenu.setItem(slotId, packet.getStateId(), item);
}
}
```

This method is responsible for all of the single slot synchronization on the client side.

Because of it, inventory changes are almost completely ignored if they occur in a non-hotbar slot (first condition) and you are in the creative mode inventory (Line 1202), but not in the "Survival Inventory" tab (Line 1203). (why?)

Potential fix:

Just remove those weird conditions. Why do they even exist in the first place?

```
public void handleContainerSetSlot(ClientboundContainerSetSlotPacket packet) {
// ...
if (packet.getContainerId() == 0) {
// ...
player.inventoryMenu.setItem(slotId, packet.getStateId(), item);
} else if (packet.getContainerId() == player.containerMenu.containerId) {
```

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-268431>